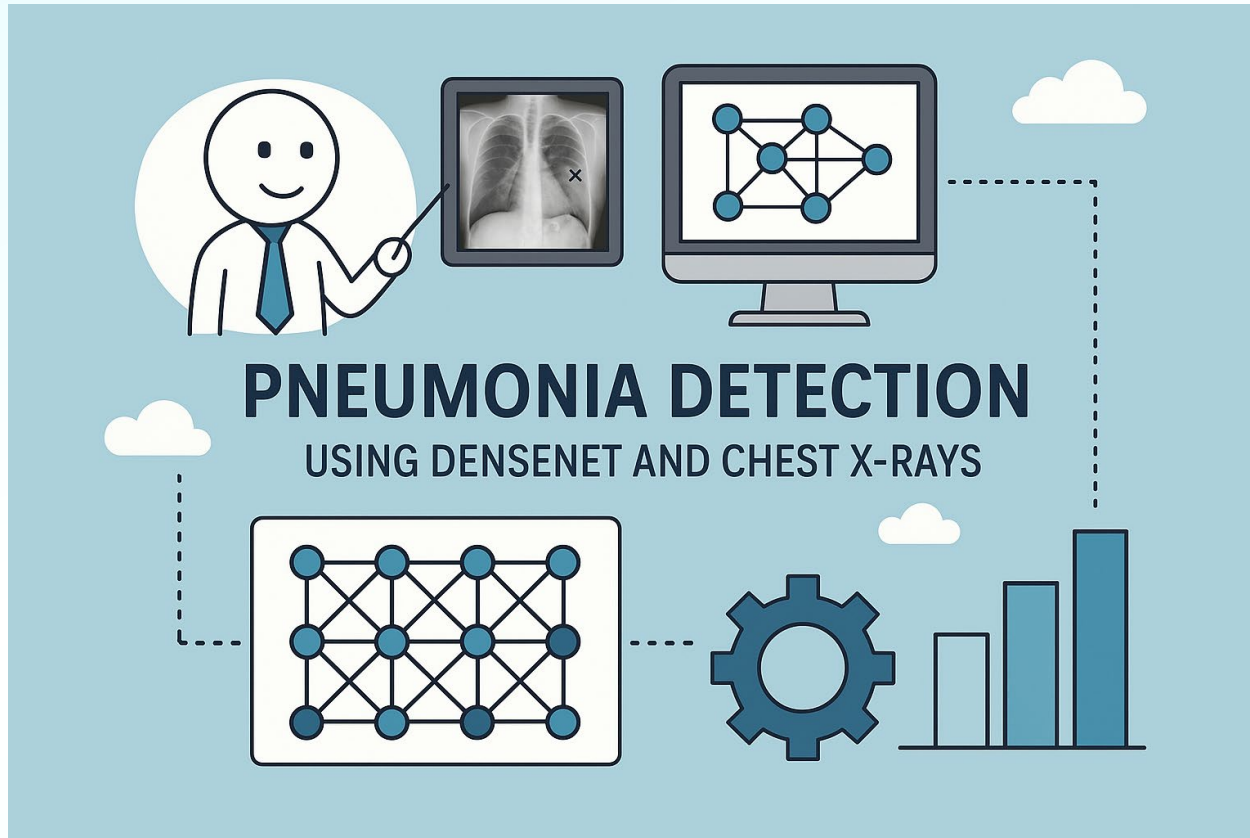




Pneumonia Detection

Presentation Outline

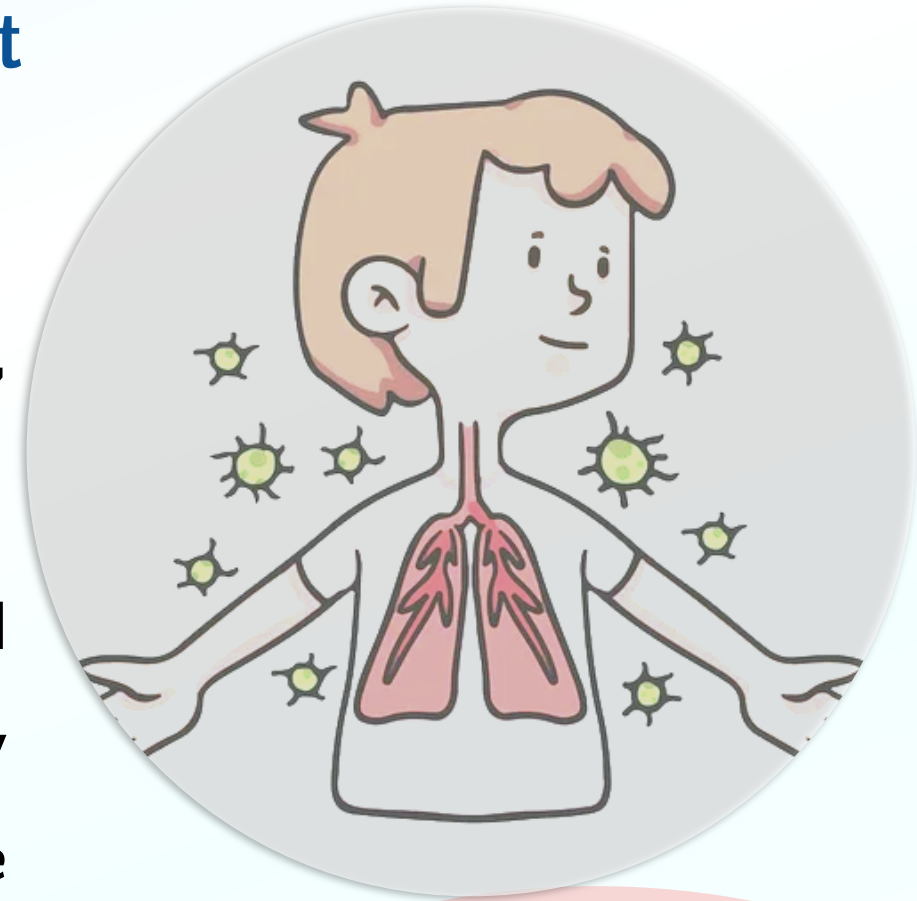


- Problem Statement
- Dataset
- Solution
- Results
- Conclusion
- Reference

Problem Statement

The Challenge of Pneumonia Detection

- Pneumonia is a significant global health concern, causing respiratory illness.
- The core problem is the need for an automated and efficient way to detect pneumonia from chest X-ray images, which can aid in faster and more accurate diagnoses compared to manual interpretation.



Dataset Overview

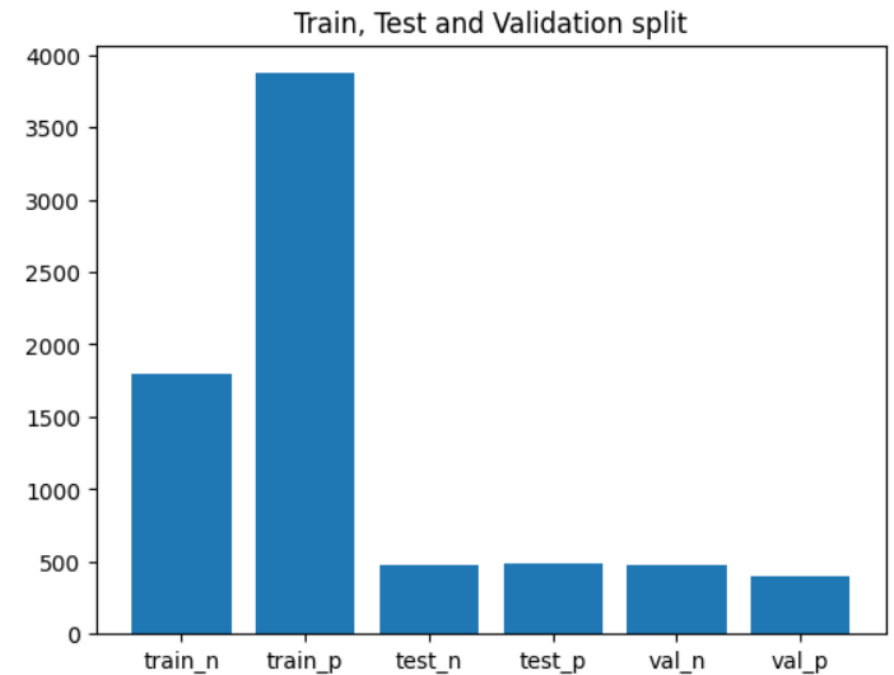
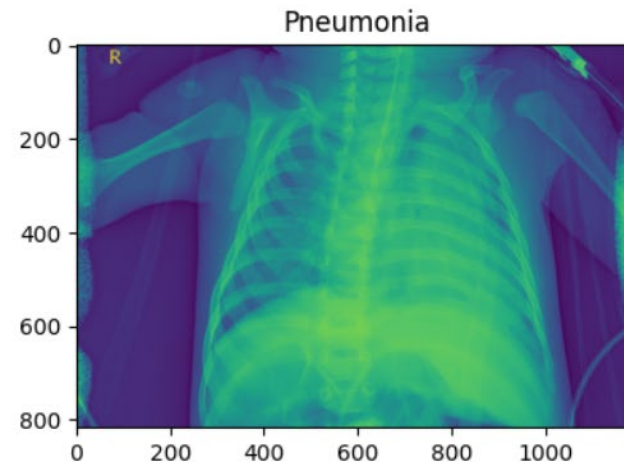
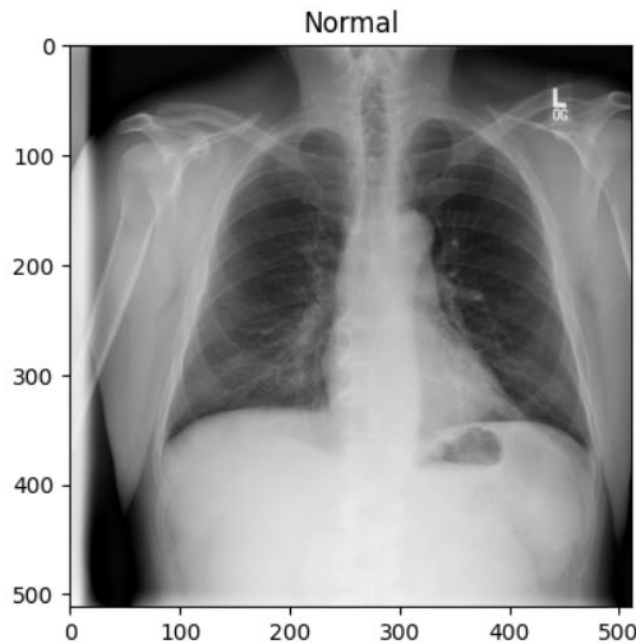
Chest X-Ray Images (Pneumonia) – Kaggle



- Pediatric X-rays (1–5 years) from Guangzhou Medical Center
- All chest X-ray imaging was performed as part of patients' routine clinical care.
- There are 6K X-Ray images (JPEG) and 2 categories (Pneumonia/Normal).

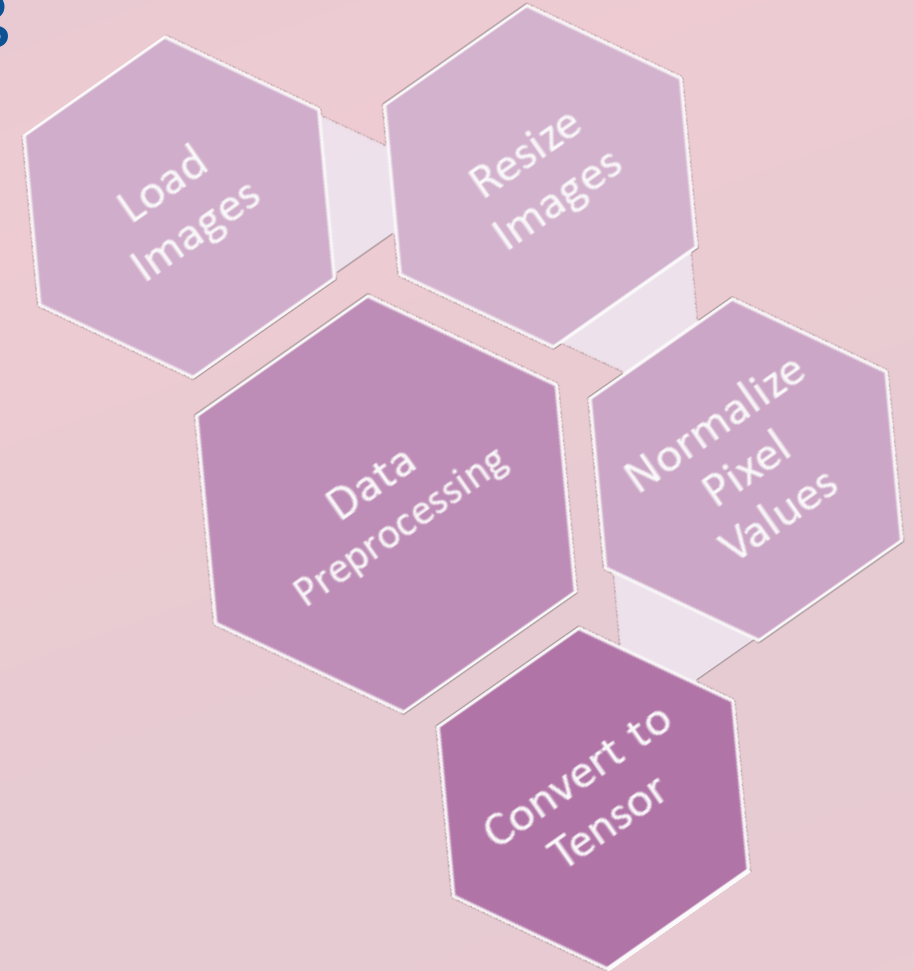
Chest X-Ray Images (Pneumonia) - Kaggle

- There are two classes in this binary classification problem: 'Normal' and 'Pneumonia'.
- The dataset is organized into three subsets: training(80%), validation(10%), and testing(10%).



Data Preprocessing

- **Resize Images:** Ensuring all images have the same dimensions.
- **Normalize Pixel Values:** Scaling pixel intensities to a specific range (e.g., 0 to 1).
- Resize to 256×256 → standard for pretrained models
- **CenterCrop** to 224×224 → matches input size of DenseNet/ResNet
- **Convert to Tensor** → shape becomes [C, H, W], normalized to [0,1]

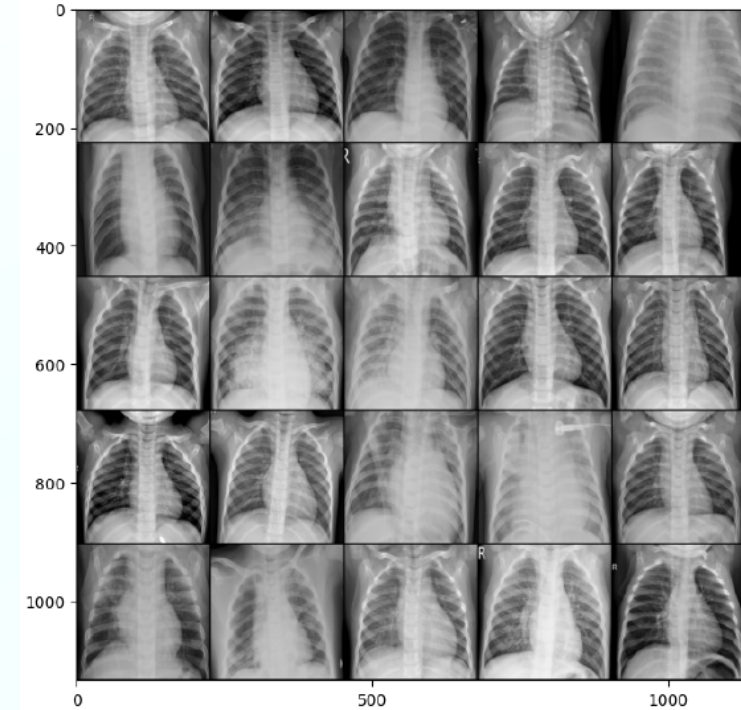


```
test_transformer = transforms.Compose([
    transforms.Resize(IMG_RESIZE),
    transforms.CenterCrop(224),
    transforms.ToTensor(),
])
```

Data Augmentation

- **RandomResizedCrop** - Forces the **model** to learn features at **different scales**.
- **RandomRotation** - Rotated the image by a 30 degrees
- **ColorJitter** - Altered the brightness(0.3), contrast(0.3) of the image.
- **HorizontalFlip**

```
train_transformer = transforms.Compose([
    transforms.RandomResizedCrop(224, scale=(0.8, 1.0)),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.3, contrast=0.3),
    transforms.RandomHorizontalFlip(),
    transforms.RandomAffine(degrees=0, translate=(0.1, 0.1)),
    transforms.RandomEqualize(),
    transforms.ToTensor(),
    AddGaussianNoise(0., 0.02),
])
```



Data Augmentation

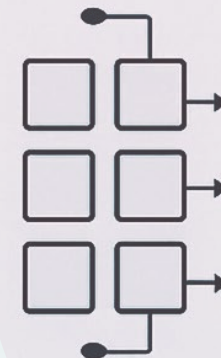
Affine Transform - Applies a combination of linear transformations to the image, including rotation, translation (shifting). transforms.

Gaussian Noise - Adds random noise to the image, following a Gaussian distribution.

Applied transforms to enhance robustness:

Data Augmentation

- Rotation
- Horizontal Flip
- Crop
- Color Jitter 
- Noise
- ...



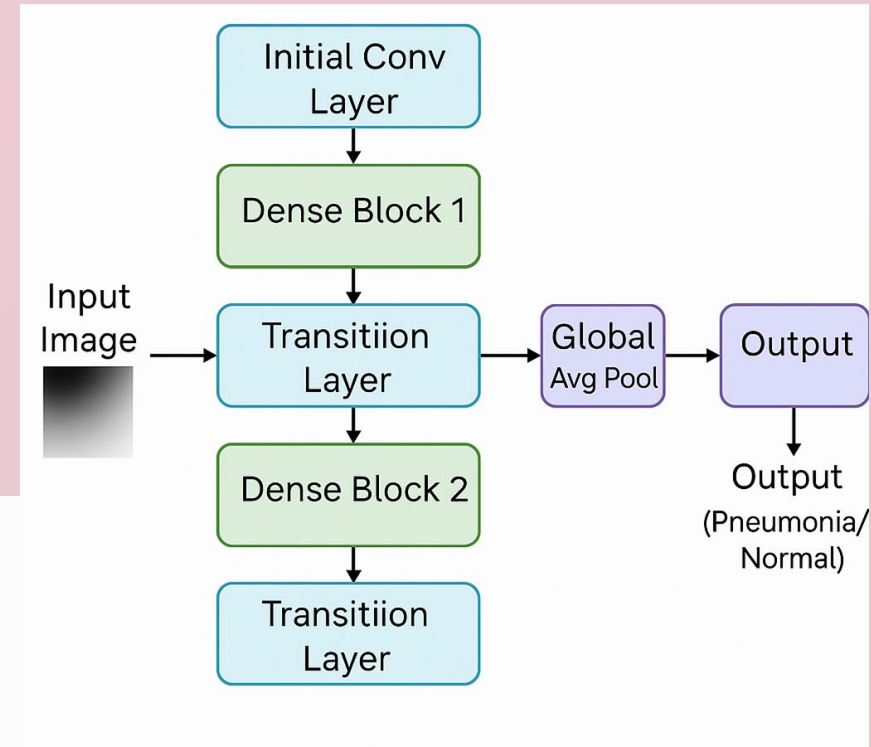
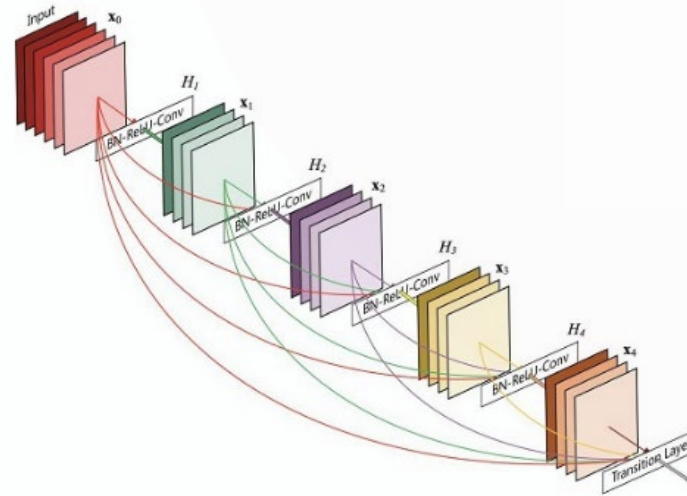
Dense layers

Why DenseNet 121?

"DenseNet" stands for Dense Convolutional Network.

The "121" refers to the number of layers in the network.

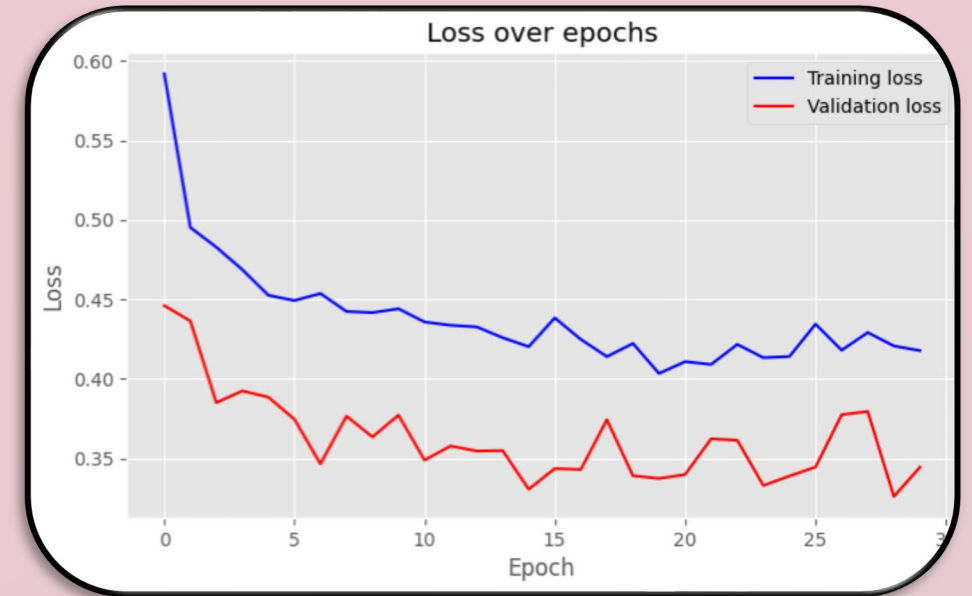
- Improved gradient flow
- Feature reuse
- Reduced parameters
- Strong performance in image classification.



Without DenseNet – pretrained(false)

```
model = torchvision.models.densenet121(pretrained=False)
```

- Started with **random weights**
- Showed **slow learning** in early epochs
- Reached **lower accuracy**
- Required **more epochs** to generalize well



```
model, optimizer, _ = training_loop(model, criterion, optimizer, train_loader, val_loader, N_EPOCHS, DEVICE)
```

Time	Epoch	Train loss	Valid loss	Train accuracy	Valid accuracy
18:17:45	0	0.5919	0.4461	83.98	83.27
18:21:47	1	0.4953	0.4365	84.20	83.63
18:25:47	2	0.4829	0.3851	86.14	90.49
18:29:48	3	0.4687	0.3925	87.17	85.92
18:33:48	4	0.4526	0.3886	87.21	89.79
18:37:48	5	0.4493	0.3747	89.13	90.67
18:41:53	6	0.4537	0.3466	88.53	90.85
18:45:52	7	0.4424	0.3766	88.53	91.20
18:49:52	8	0.4417	0.3636	89.72	89.79
18:53:51	9	0.4441	0.3773	90.66	89.96
18:57:49	10	0.4358	0.3490	90.88	91.20
19:02:01	11	0.4337	0.3579	91.15	88.20

DenseNet – Transfer Learning

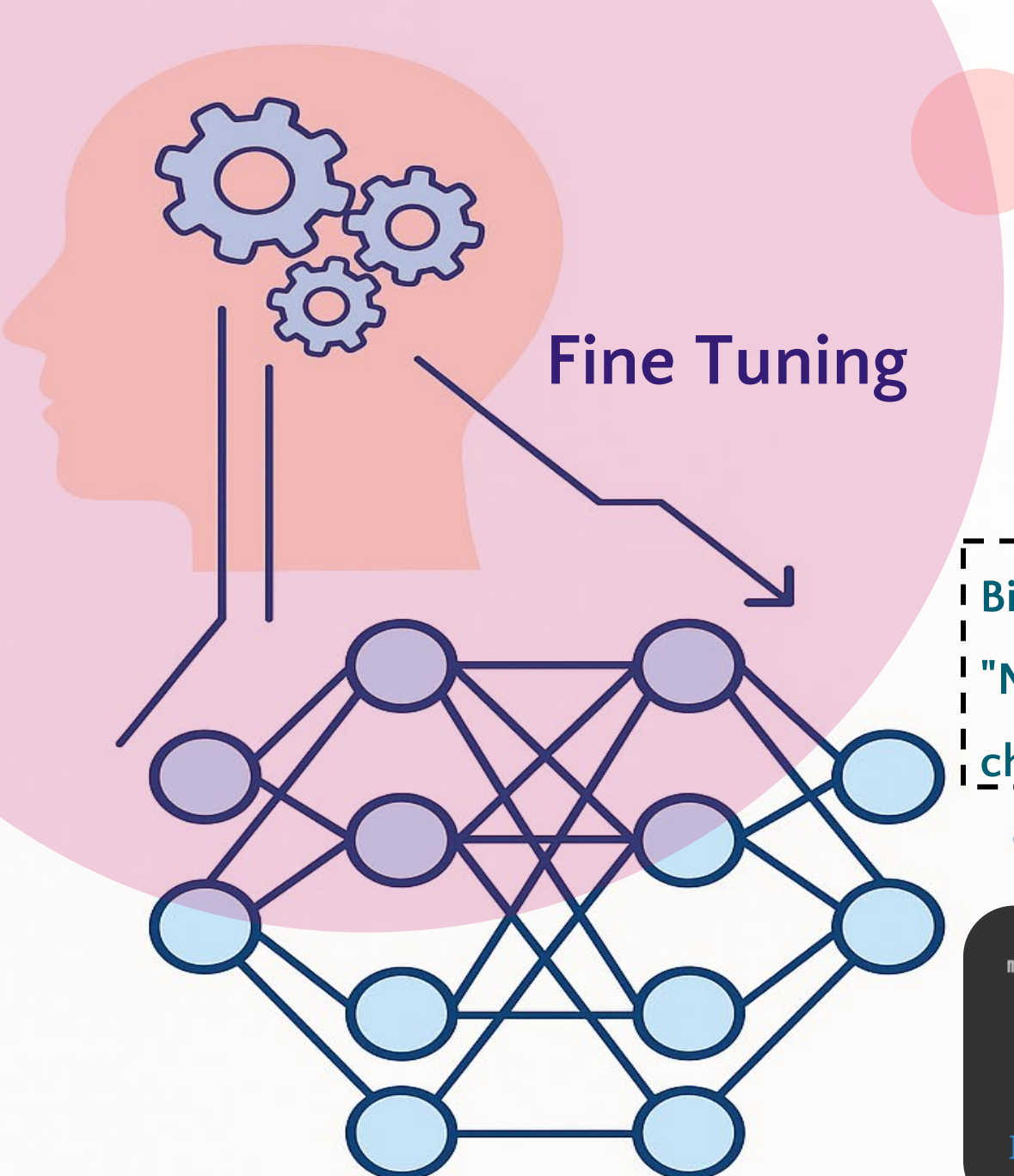
```
model = torchvision.models.densenet121(pretrained=True)
```

- Pretrained DenseNet121
- Final layer replaced with a custom 2-class head
- Data efficiency
- Faster training
- Improved generalization



```
model, optimizer, _ = training_loop(model, criterion, optimizer, train_loader, val_loader, N_EPOCHS, DEVICE)
```

```
08:24:49 --- Epoch: 0   Train loss: 0.4337   Valid loss: 0.3337   Train accuracy: 94.03   Valid accuracy: 94.37
08:28:55 --- Epoch: 1   Train loss: 0.3740   Valid loss: 0.3097   Train accuracy: 94.69   Valid accuracy: 92.25
08:32:55 --- Epoch: 2   Train loss: 0.3650   Valid loss: 0.3146   Train accuracy: 94.26   Valid accuracy: 92.25
08:36:55 --- Epoch: 3   Train loss: 0.3601   Valid loss: 0.3098   Train accuracy: 94.32   Valid accuracy: 93.49
08:40:56 --- Epoch: 4   Train loss: 0.3569   Valid loss: 0.3245   Train accuracy: 95.20   Valid accuracy: 93.13
08:44:58 --- Epoch: 5   Train loss: 0.3448   Valid loss: 0.2935   Train accuracy: 95.54   Valid accuracy: 94.72
08:48:58 --- Epoch: 6   Train loss: 0.3433   Valid loss: 0.3164   Train accuracy: 95.26   Valid accuracy: 94.89
08:52:58 --- Epoch: 7   Train loss: 0.3432   Valid loss: 0.3146   Train accuracy: 95.50   Valid accuracy: 95.77
08:57:00 --- Epoch: 8   Train loss: 0.3356   Valid loss: 0.2958   Train accuracy: 96.48   Valid accuracy: 94.01
09:01:03 --- Epoch: 9   Train loss: 0.3342   Valid loss: 0.3224   Train accuracy: 95.77   Valid accuracy: 94.89
Early stopping triggered.
```

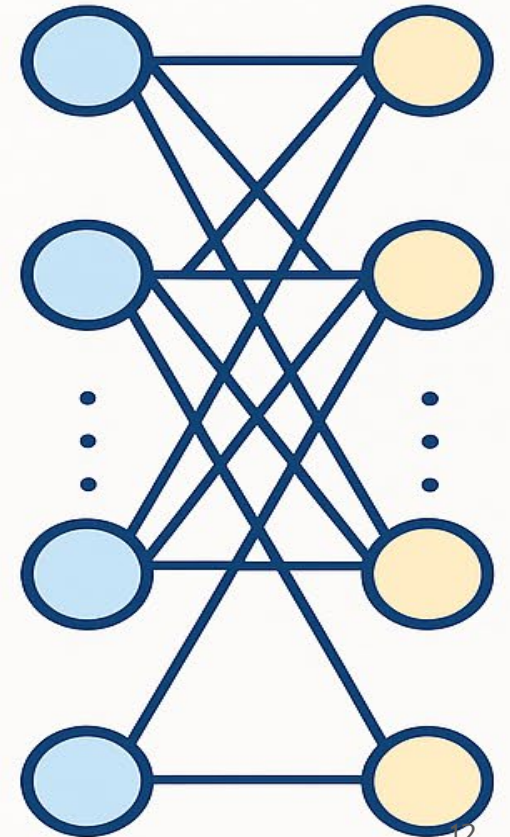


Fine Tuning

The final classification layer of the DenseNet-121 model, which is originally designed for 1000 classes in ImageNet, is replaced with a new linear layer.

Binary classification task :
"Normal" and "Pneumonia" in
chest X-rays.

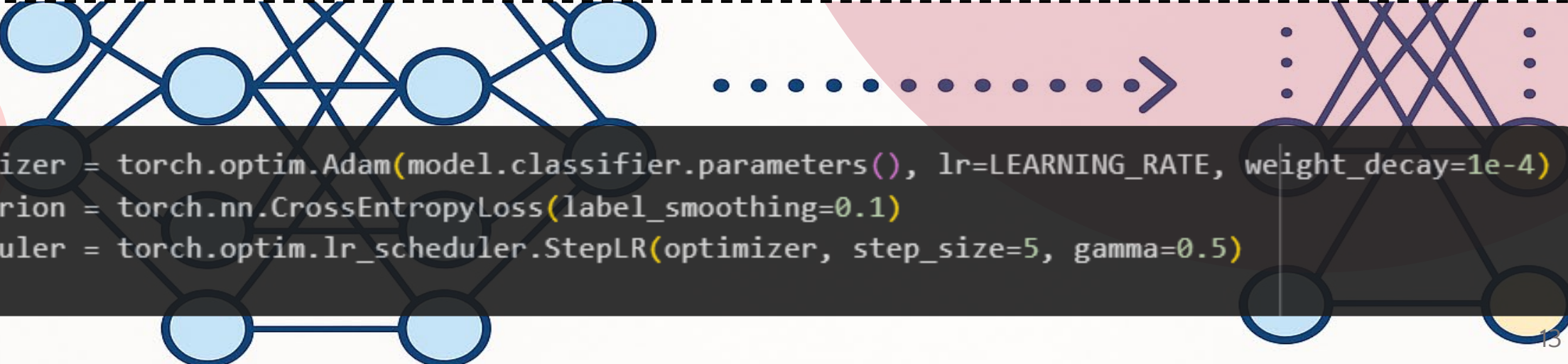
```
model.classifier = nn.Sequential(OrderedDict([  
    ('fc1', nn.Linear(in_features, 64)),  
    ('relu1', nn.ReLU()),  
    ('drop1', nn.Dropout(0.5)),  
    ('fc2', nn.Linear(64, 2))  
]))
```



Fine Tuning



- An optimizer (Adam) is used to adjust the model's weights based on the calculated loss.
- The loss function (Cross-Entropy Loss) measures the difference between the model's predictions and the actual labels.
- By updating the weights through backpropagation, the model learns to map chest X-ray images to the correct class (Normal or Pneumonia).



```
optimizer = torch.optim.Adam(model.classifier.parameters(), lr=LEARNING_RATE, weight_decay=1e-4)
criterion = torch.nn.CrossEntropyLoss(label_smoothing=0.1)
scheduler = torch.optim.lr_scheduler.StepLR(optimizer, step_size=5, gamma=0.5)
```

Training Configuration

Loss: CrossEntropyLoss

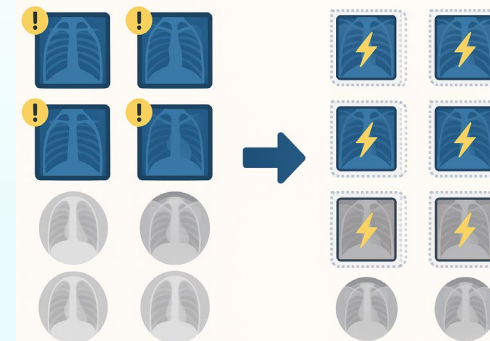
Optimizer: Adam, lr = 1e-4

Epochs: 30

Batch Size: 32

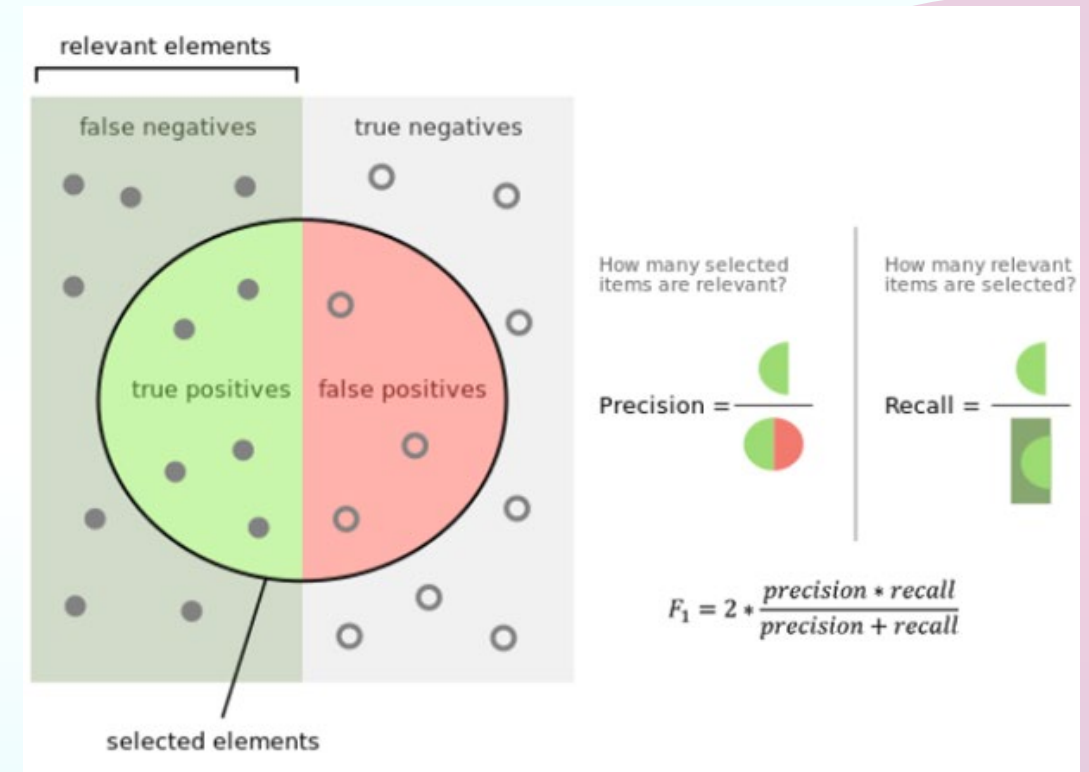
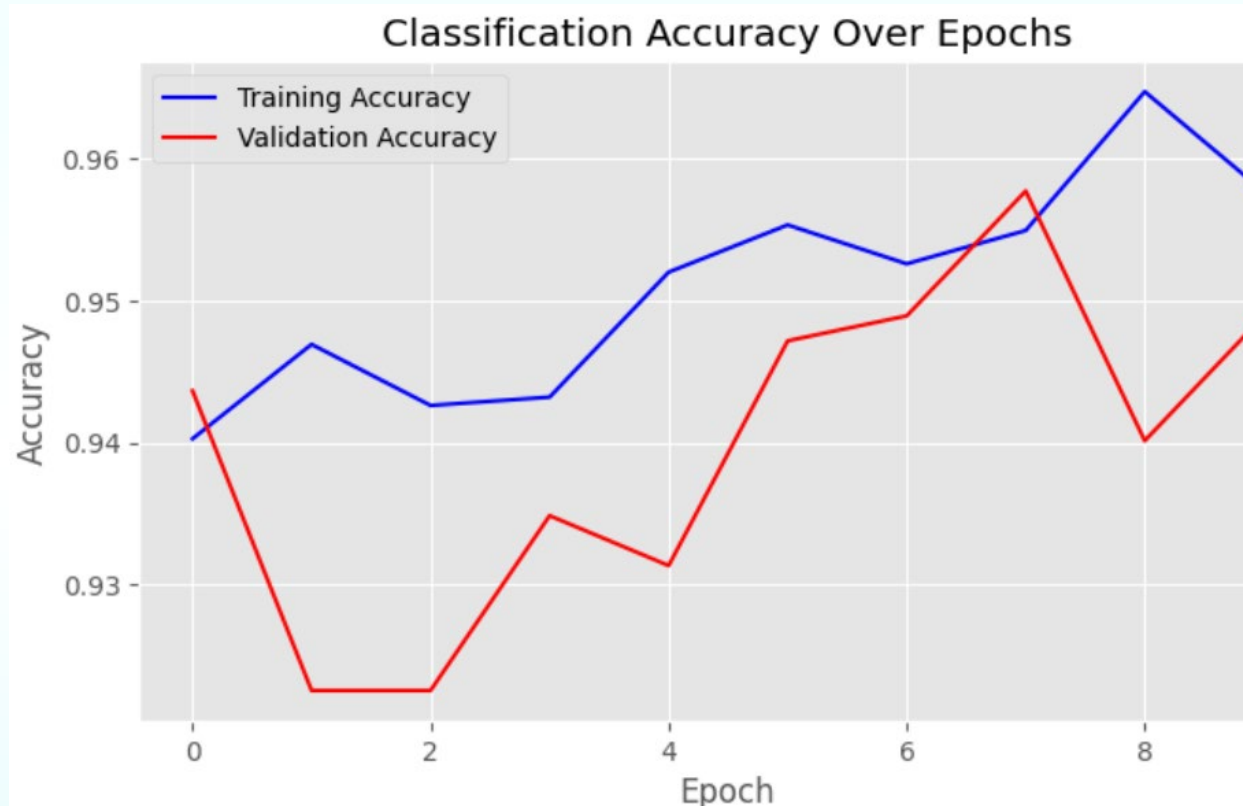
```
from sklearn.model_selection import train_test_split
random_seed = 0
BATCH_SIZE = 32
IMG_RESIZE = (256, 256)
torch.manual_seed(random_seed)
```

```
LEARNING_RATE = 0.001
N_EPOCHS = 30
N_CLASSES = 2
DEVICE = 'cuda:0' if torch.cuda.is_available() else 'cpu'
```

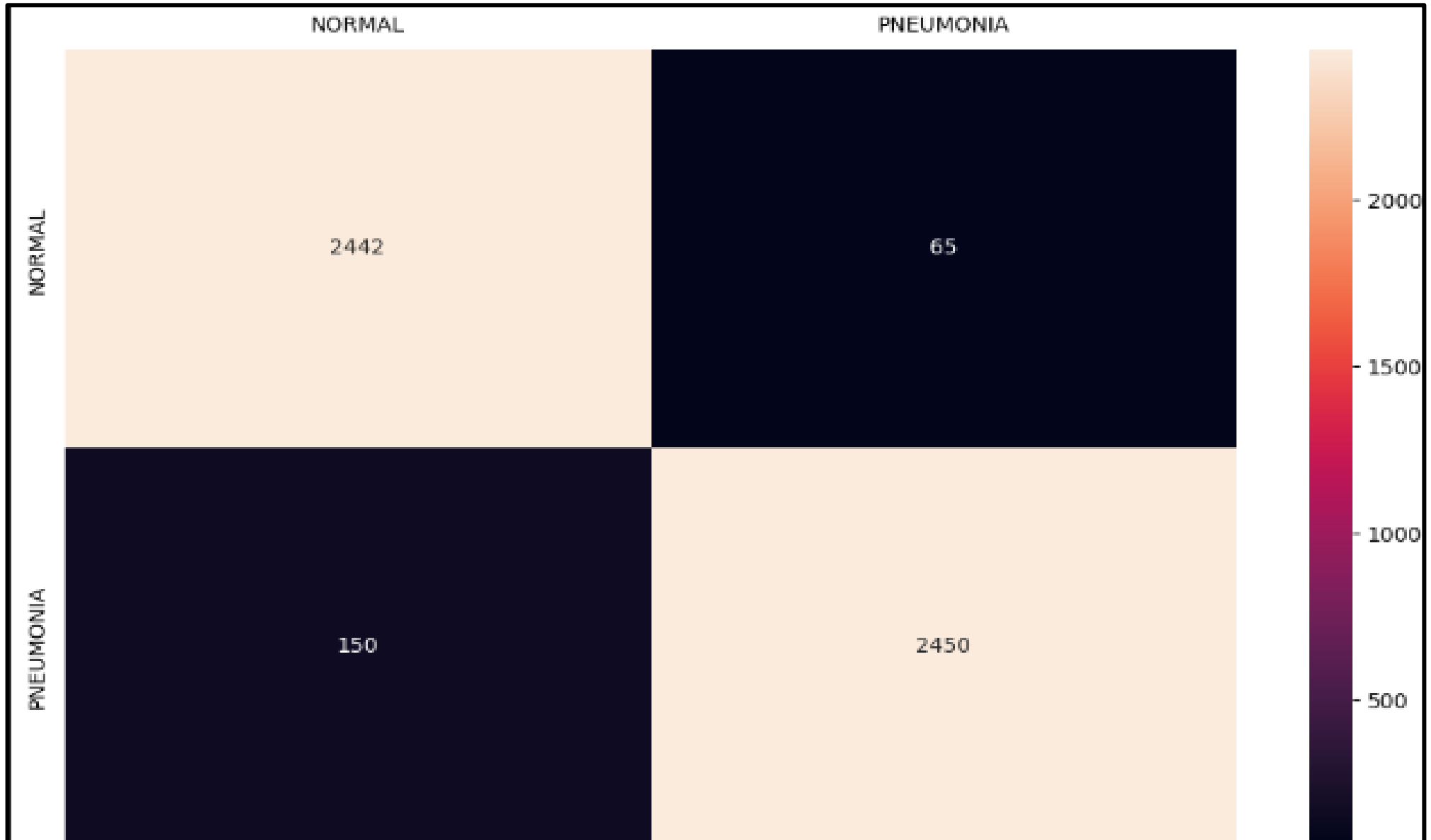


Evaluation Metrics

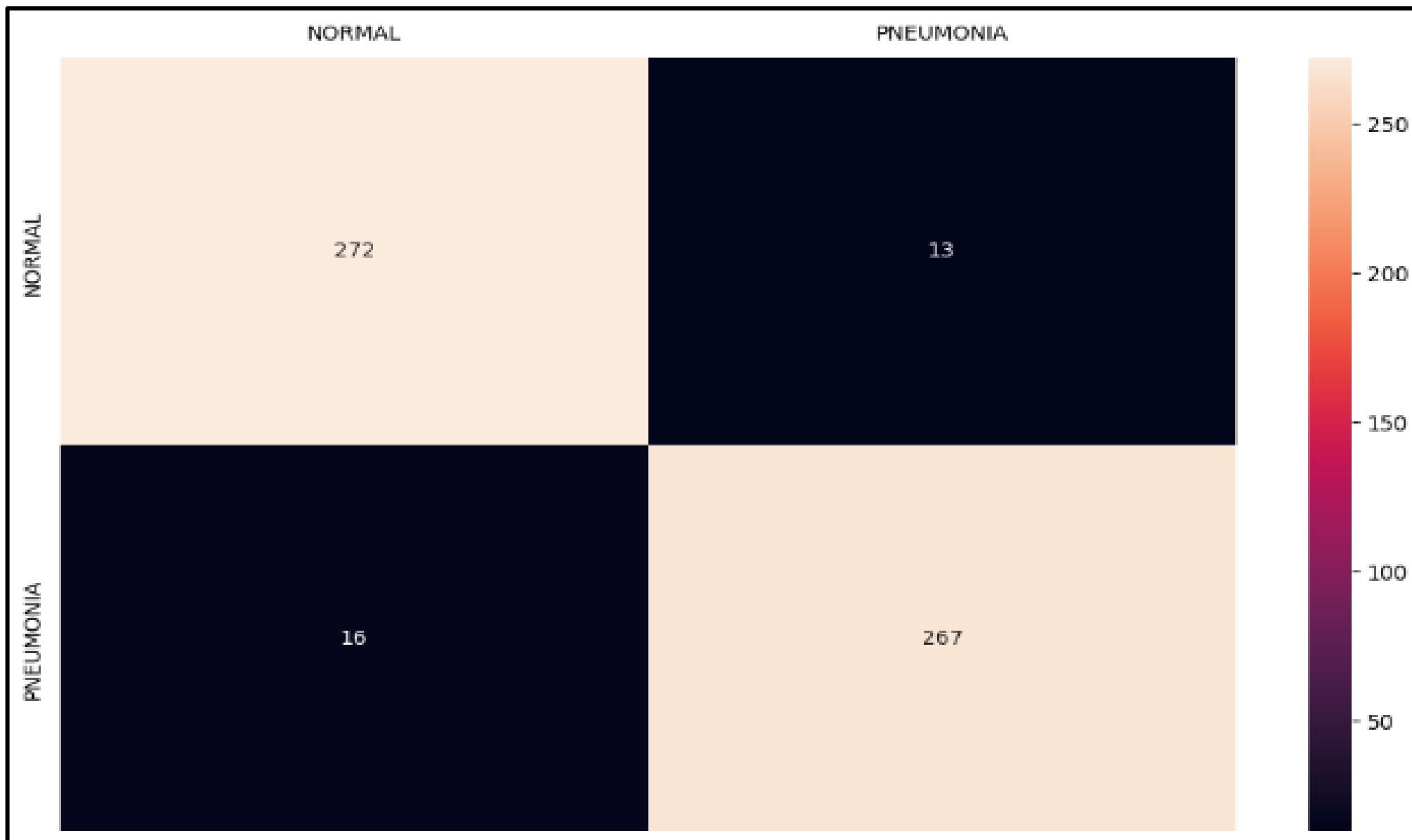
Accuracy, Precision, Recall, F1-Score, ROC Curve, AUC



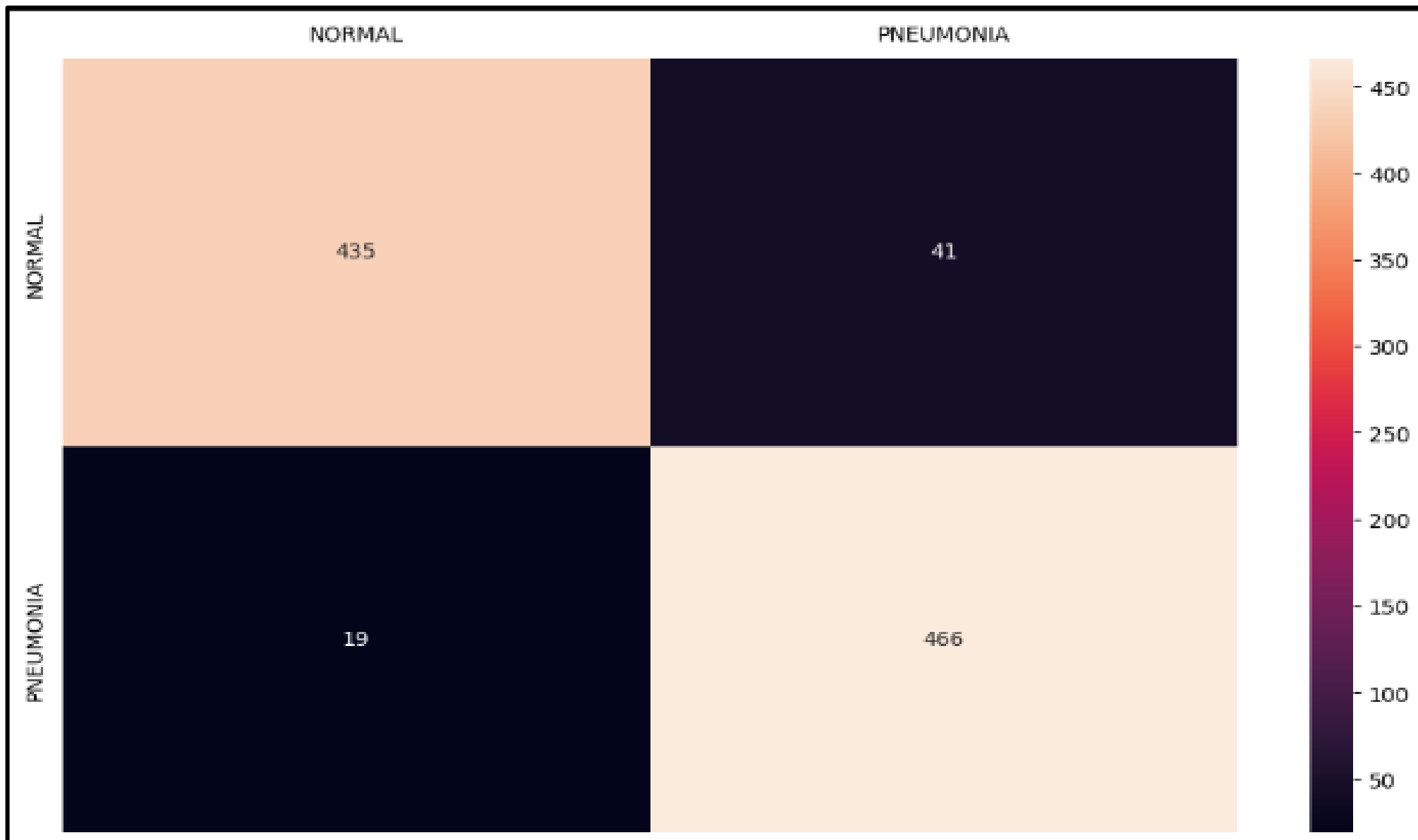
Train set result



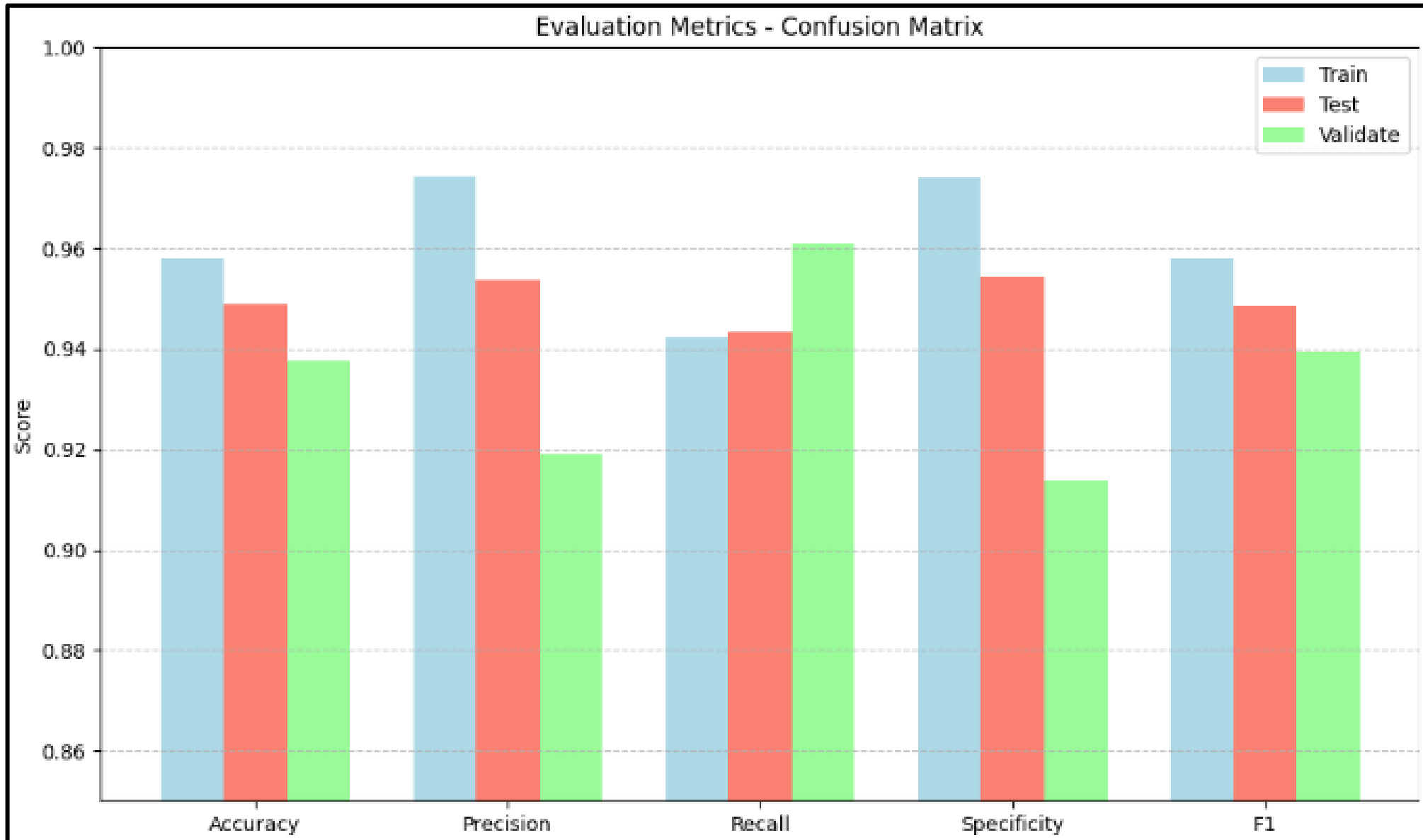
Test set result



Validation set result



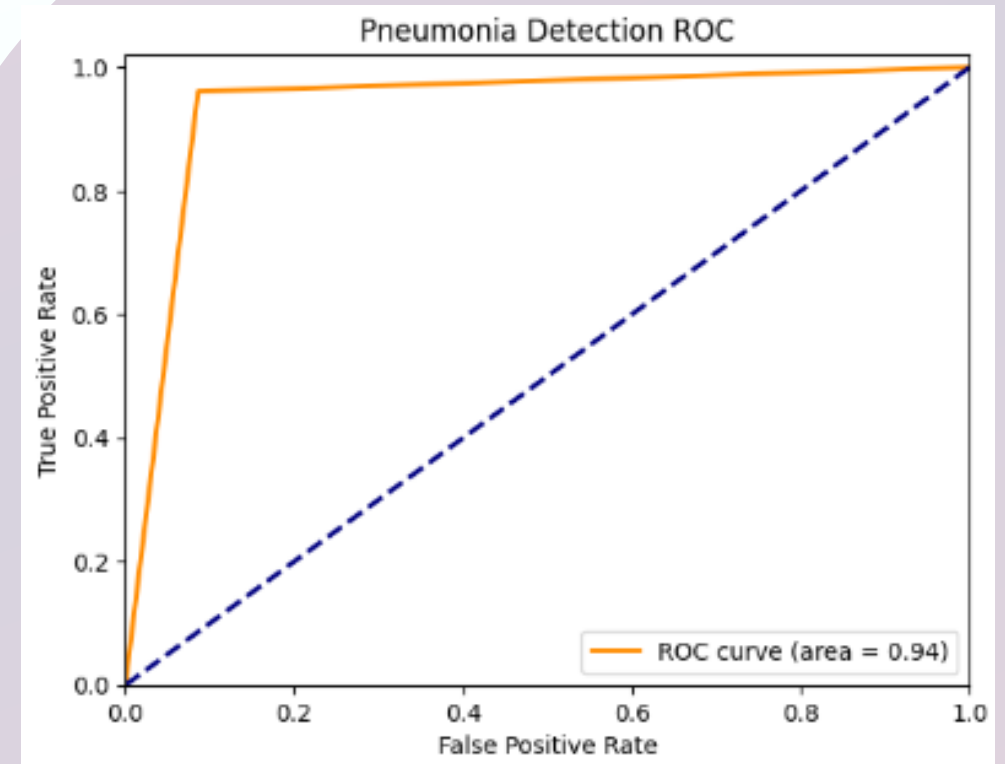
Confusion matrix - comparison



Result

- The DenseNet-121 model achieved a test set **accuracy** of **93%**.
- The model's **precision** and **recall** for pneumonia detection were **91%** and **96%**, respectively, with an **F1-score** of **93%**.
- The AUC-ROC curve demonstrated strong discriminatory power, with an AUC of [0.94].

Pneumonia Detection from Chest X-ray Images using DenseNet



Conclusion

- The DenseNet-121 model, fine-tuned using transfer learning, effectively distinguishes between normal and pneumonia-affected chest X-ray images.
- Transfer learning proves to be a valuable technique for medical image analysis, enabling good performance with limited data.



Future Work

- **Dataset Expansion:**

Increase the size and diversity of the dataset to improve the model's generalization ability.

Include data from various sources and patient demographics.



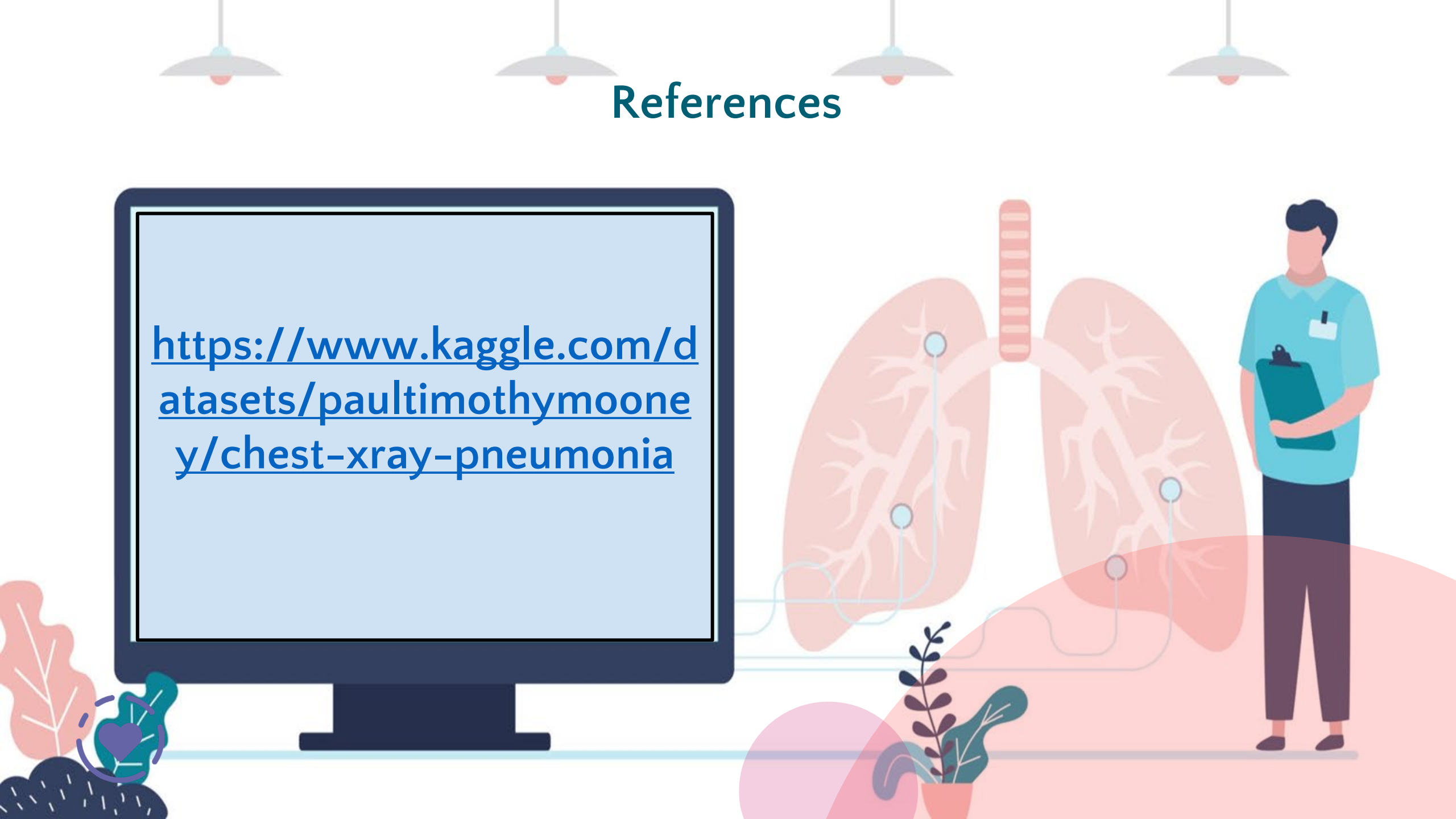
- **Model Enhancement:**

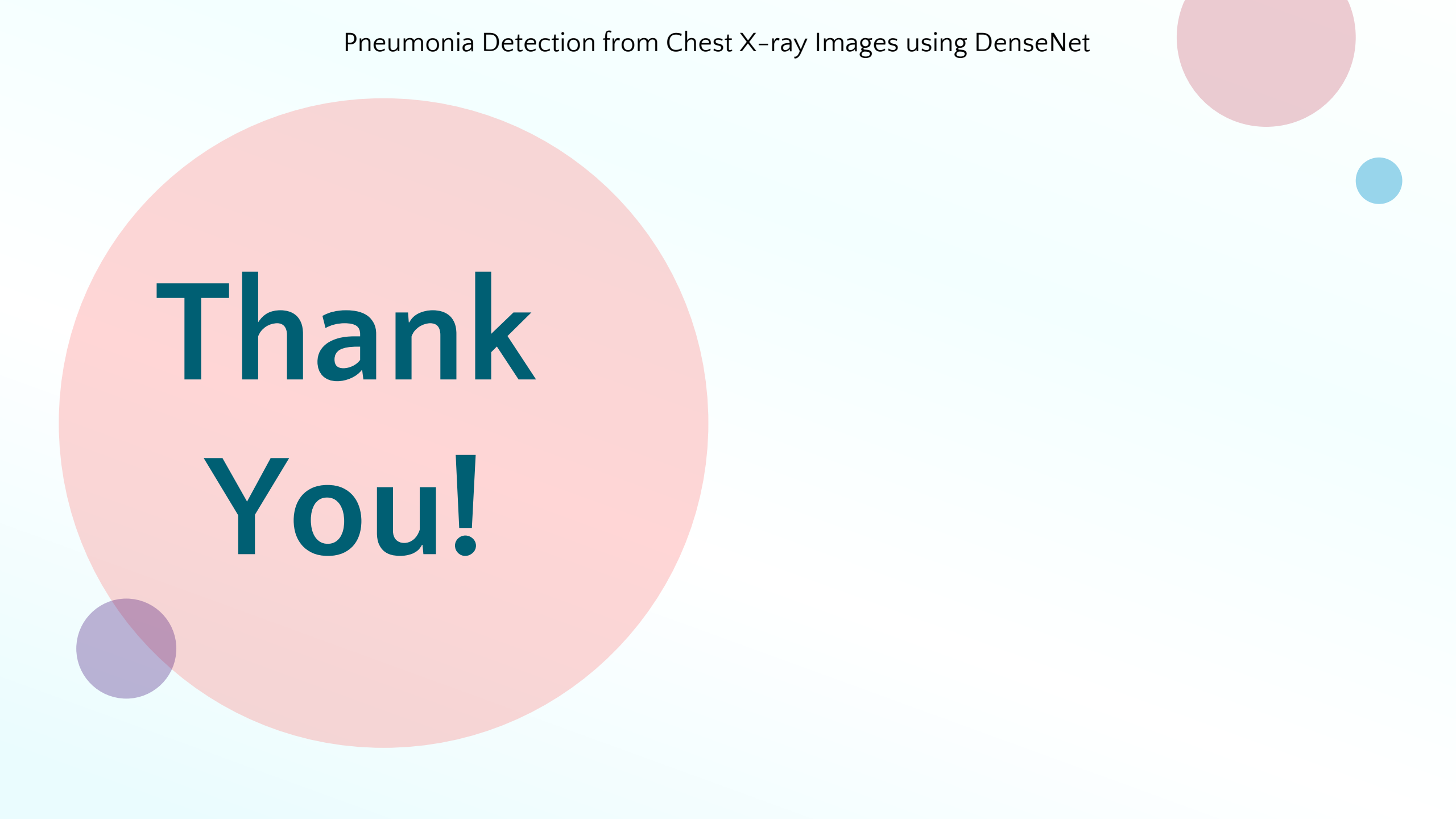
Explore attention mechanisms to help the model focus on relevant regions in the X-ray images.



References

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>





Thank You!